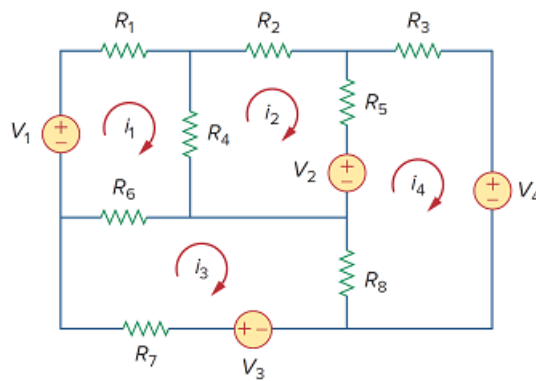


Problem

By inspection, obtain the mesh-current equations for the circuit in Fig. 3.118.

Figure 3.118:



Step-by-step solution

Step 1 of 5

Refer to Figure 3.118 in the textbook.

There are four meshes, so the resistance matrix is 4 by 4.

The diagonal terms of the matrix are,

$$R_{11} = R_1 + R_4 + R_6$$

$$R_{22} = R_2 + R_5 + R_4$$

$$R_{33} = R_6 + R_7 + R_8$$

$$R_{44} = R_3 + R_5 + R_8$$

The off-diagonal terms of the matrix from loop1 are,

$$R_{12} = R_4$$

$$R_{13} = R_6$$

$$R_{14} = 0$$

Step 2 of 5

The off-diagonal terms of the matrix from loop2 are,

$$R_{21} = -R_4$$

$$R_{23} = 0$$

$$R_{24} = -R_5$$

The off-diagonal terms of the matrix from loop3 are,

$$R_{31} = -R_6$$

$$R_{32} = 0$$

$$R_{34} = -R_8$$

The off-diagonal terms of the matrix from loop4 are,

$$R_{41} = 0$$

$$R_{42} = -R_5$$

$$R_{43} = -R_8$$

Step 3 of 5

The resistance matrix is,

$$\mathbf{R} = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ R_{21} & R_{22} & R_{23} & R_{24} \\ R_{31} & R_{32} & R_{33} & R_{34} \\ R_{41} & R_{42} & R_{43} & R_{44} \end{bmatrix}$$

$$= \begin{bmatrix} R_1 + R_4 + R_6 & -R_4 & -R_6 & 0 \\ -R_4 & R_2 + R_5 + R_4 & 0 & -R_5 \\ -R_6 & 0 & R_6 + R_7 + R_8 & -R_8 \\ 0 & -R_5 & -R_8 & R_3 + R_5 + R_8 \end{bmatrix}$$

Step 4 of 5

The input voltage terms of vector $\mathbf{v}$ are,

$$v_1 = V_1$$

$$v_2 = -V_2$$

$$v_3 = V_3$$

$$v_4 = V_2 - V_4$$

The input voltage vector $\mathbf{v}$ is,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

$$= \begin{bmatrix} V_1 \\ -V_2 \\ V_3 \\ V_2 - V_4 \end{bmatrix}$$

Step 5 of 5

Write the mesh current equations in matrix form.

$$\mathbf{R}\mathbf{i} = \mathbf{v}$$

$$\begin{bmatrix} R_1 + R_4 + R_6 & -R_4 & -R_6 & 0 \\ -R_4 & R_2 + R_5 + R_4 & 0 & -R_5 \\ -R_6 & 0 & R_6 + R_7 + R_8 & -R_8 \\ 0 & -R_5 & -R_8 & R_3 + R_5 + R_8 \end{bmatrix} \mathbf{i} = \begin{bmatrix} V_1 \\ -V_2 \\ V_3 \\ V_2 - V_4 \end{bmatrix}$$

Thus, the mesh current equations in matrix form is,

$$\boxed{\begin{bmatrix} R_1 + R_4 + R_6 & -R_4 & -R_6 & 0 \\ -R_4 & R_2 + R_5 + R_4 & 0 & -R_5 \\ -R_6 & 0 & R_6 + R_7 + R_8 & -R_8 \\ 0 & -R_5 & -R_8 & R_3 + R_5 + R_8 \end{bmatrix} \mathbf{i} = \begin{bmatrix} V_1 \\ -V_2 \\ V_3 \\ V_2 - V_4 \end{bmatrix}}$$